



Analog IC Design Implementation and Verification
Week 1 – Introduction to Analog IC Design and Processes

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Module Objectives

In this module, you

- Learn how to use NMOS and PMOS transistors to create basic circuit designs and analyze designs using mathematical models of transistor behavior
- Explain the current and voltage characteristics of MOSFETs
- Use the different functional models to describe the performance of MOSFETs
- Explain the steps in the fabrication process for CMOS devices
- Explain the parameters that affect the switching performance of CMOS logic gates
- Describe the differences among various kinds of logic gates



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Introduction: MOSFETs Characteristics

- How does a MOSFET (metal-oxide-semiconductor field effect transistor) work?
- Current and voltage (I/V) characteristics
- Second-order effects

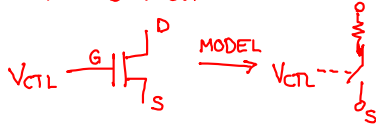


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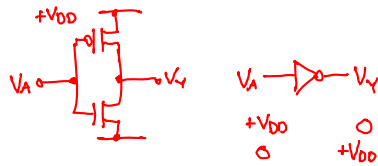
Introduction: MOSFETs Applications

MOSFET FUNCTIONS

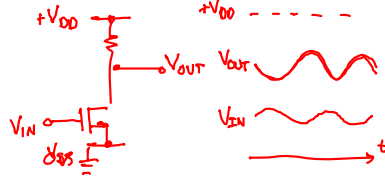
ANALOG SWITCH



DIGITAL SWITCH



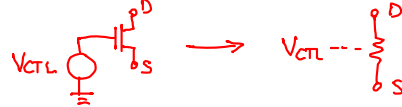
AMPLIFIER



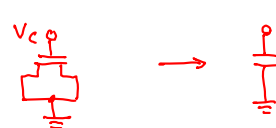
CURRENT SOURCE



RESISTOR



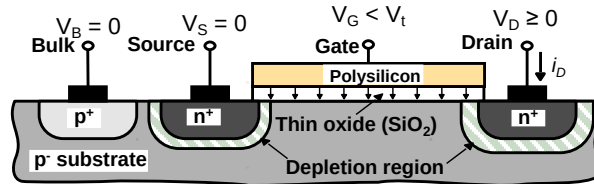
CAPACITOR



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MOSFET Modes: Cutoff

- How does a MOSFET work? Assume the MOSFET is an enhancement type. The first mode is called the *cutoff mode*, and in this mode, the channel is not yet formed. (Actually, a small current that is called a *subthreshold current* flows but can be ignored for now.)
- Electric-field strength is shown by the spacing of the arrows under the polysilicon layer.

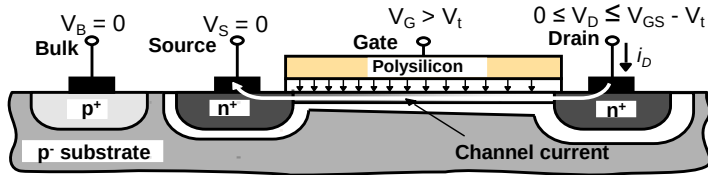


NMOS (n-channel metal oxide semiconductor) transistor

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MOSFET Modes: Active

- The second mode is called the *active mode* (also known as nonsaturated, linear, or triode mode).
- The region under the gate is inverted and current flows from drain to source.

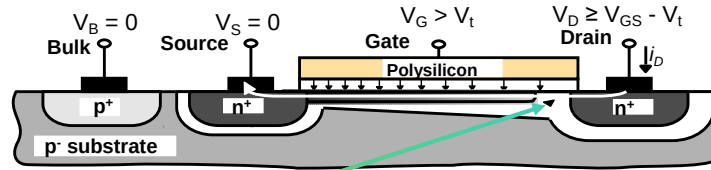


The arrow separation in the thin oxide represents the electric field strength.

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MOSFET Modes: Saturation

- The third mode is called the *saturation mode* (also known as the pentode mode). The drain voltage has increased to the point where a depletion region is formed between the channel and the drain. The drain voltage loses control over the drain current.



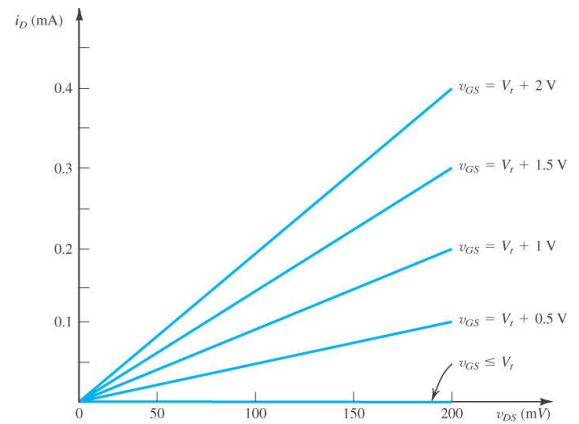
A depletion region forms between the drain and channel

The arrow separation in the thin oxide represents the electric-field strength.

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I/V Characteristics for Low v_{DS} (Triode)

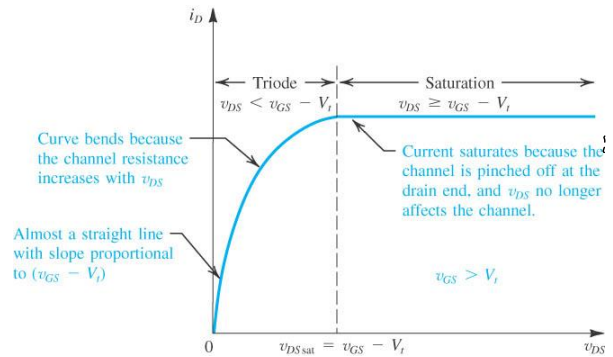
- i_D is a function of the gate voltage.
- When $v_{GS} \leq V_t$, no current flows because no channel is induced.
- As v_{GS} increases, i_D increases for a fixed v_{DS} .
- For a fixed v_{GS} , i_D is proportional to v_{DS} . Therefore, the MOSFET acts like a variable resistor.



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I/V Characteristics for Larger v_{DS}

- For low v_{DS} , the I/V curve is almost linear and acts like a resistor.
- As v_{DS} approaches v_{DSsat} , the curve goes nonlinear.
- Once v_{DS} exceeds v_{DSsat} , the depletion region is formed, and the current is no longer a function of v_{DS}



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Derivation of the I/V Relationship (continued)

First calculate the capacitance of a strip dx wide:

$$C_{strip} = C_{ox} W dx \quad \text{where } C_{ox} \text{ is the capacitance per unit gate area}$$

Because $q = CV$,

$$dq = -C_{ox} W dx [(v_{GS} - V_t) - v(x)]$$

$$\therefore \frac{dq}{dx} = -C_{ox} W [(v_{GS} - V_t) - v(x)] \quad \text{with the change in sign due to } dq \text{ being a negative charge.}$$

The ultimate goal is to find i as a function of $v(x)$, so

$$i = \frac{dq}{dt} = \frac{dq}{dx} \frac{dx}{dt}$$

You can get dq/dx from the second set of equations, so you need to find dx/dt :

$$E(x) = -\frac{dv(x)}{dx}$$

$$\frac{dx}{dt} = -\mu_n E(x) = \mu_n \frac{dv(x)}{dx} \quad \text{where } \mu_n \text{ is the mobility of the electrons in the channel.}$$



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Derivation of the I/V Relationship (continued)

Now solve for i_D , which is the drain current and the opposite of i :

$$i_D = -i = -\frac{dq}{dx} \frac{dx}{dt} = C_{ox} W [(v_{GS} - V_t) - v(x)] \cdot \mu_n \frac{dv(x)}{dx}$$

$$\therefore i_D dx = \mu_n C_{ox} W [(v_{GS} - V_t) - v(x)] dv(x)$$

Integrating both sides and rearranging:

$$\int_0^L i_D dx = \int_{v(0)=0}^{v(L)=v_{DS}} \mu_n C_{ox} W [(v_{GS} - V_t) - v(x)] dv(x)$$

$$i_D L = \mu_n C_{ox} W \int_{v(0)=0}^{v(L)=v_{DS}} (v_{GS} - V_t) dv(x) - \mu_n C_{ox} W \int_{v(0)=0}^{v(L)=v_{DS}} v(x) dv(x)$$

$$\therefore i_D = \mu_n C_{ox} \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right] \quad (\text{triode region})$$

$$\therefore i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 \quad (\text{saturation region, where } v_{DS} = v_{GS} - V_t)$$



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Second-Order Effects

The i_D equation from the previous slide gives a good first-order approximation to the performance of a MOSFET. However, secondary effects have to be accounted for.

Some second-order effects include the following:

- Body effect
- Channel-length modulation
- Subthreshold conduction



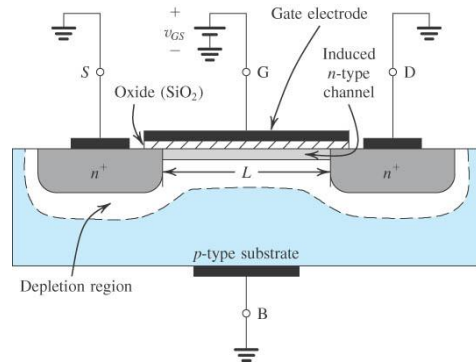
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Body Effect

Definition: The effect on MOSFET performance due to the voltage difference between the source and the substrate.

- Usually, the substrate is tied to the lowest voltage (for NMOS) to avoid the forward biasing of any pn junctions.
- The more reverse-biased the pn junction becomes, the larger the depletion region grows, which reduces the channel depth.
- When the channel depth is reduced, more v_{GS} is required to restore it to its previous depth.
- The body effect is usually accounted for by modifying the equation for the threshold voltage V_t :

$$V_t = V_{t0} + \gamma \left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f} \right]$$



- V_{t0} is the threshold voltage for $V_{SB} = 0$.
- Φ_f is a physical parameter approximately equal to 0.3V.
- γ is a fabrication-process parameter typically around 0.4√V.

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Channel-Length Modulation (continued)

Making a substitution for ΔL
(and assuming that ΔL is
proportional to v_{DS}):

$$\Delta L = \lambda' v_{DS}$$

Substituting this into the
previous equation gives

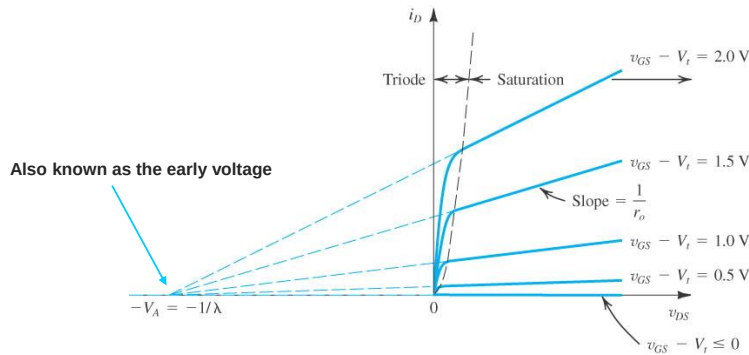
$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) \left(1 + \frac{\lambda'}{L} v_{DS} \right) (v_{GS} - V_t)^2$$

Making another substitution

$$\lambda = \frac{\lambda'}{L}$$

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 (1 + \lambda v_{DS})$$

where λ is a process-technology parameter with
units of V^{-1}



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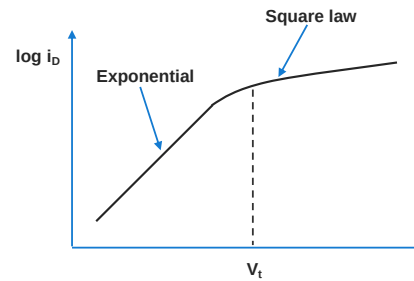
Subthreshold Conduction

Definition: The conduction of current from drain to source even though v_{GS} is below V_t .

- To a first order, there is no drain current when v_{GS} is below V_t .
- To a second order, however, this is not strictly true.
- There is an exponential relationship between this subthreshold current and v_{GS} :

$$i_D = i_0 e^{\frac{v_{GS}}{\zeta V_t}}$$

where $\zeta > 1$ is a nonideality factor.



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Summary: MOSFET Characteristics

- Three MOSFET modes
 - Cutoff
 - Active (nonsaturated, linear, triode)
 - Saturation (pentode)
- MOSFET I/V characteristic
 - Different equations for different modes
 - Derivation of the drain current equation for subthreshold and saturation regions
- Second-order effects
 - Body effect
 - Channel-length modulation
 - Subthreshold conduction



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Introduction: MOSFET Device Models

- Models provide abstraction from the physics of the device.
- Simplified models are useful for basic understanding and hand calculations. Computer models can be far more complex with hundreds of possible parameters.
- Models should be as simple as possible for hand calculations.
- Simulation programs also use simplifications of models where appropriate (for example, AC analysis versus transient analysis).
- The main industry focus is now on very sophisticated models that enable covering the whole range from linear analysis up to the inclusion of second- and third-order effects.



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Characterization of Models

		Time dependence	
		Static	Dynamic
Linearity	Linear	Small-signal, midband: R_{in}, A_v, R_{out}	Small-signal frequency response: poles and zeros C_{gs}, C_{gd}
	Nonlinear	DC operating point: $i_D = f(V_D, V_G, V_S, V_B)$	Large-signal transient response: slew rate

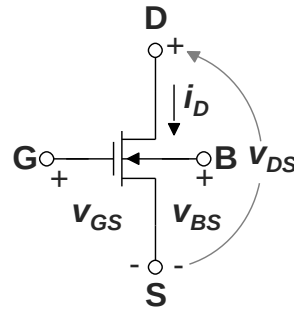


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MOSFET Large-Signal Models

The three regions of operation include:

- Subthreshold
 - $v_{GS} \leq V_t$
 - Channel not yet fully formed
- Active (nonsaturation)
 - $v_{DS} < v_{GS} - V_t$
 - Channel exists; current depends on v_{DS} and v_{GS} .
- Saturation
 - $v_{DS} > v_{GS} - V_t$
 - Channel exists; current primarily a function of v_{GS} .



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First-Order Large-Signal Model Equations

- Subthreshold ($v_{GS} \leq V_t$)

$$i_D = i_0 e^{\frac{v_{GS}}{V_t}}$$

- Nonsaturation ($v_{DS} < v_{GS} - V_t$)

$$i_D = \frac{W\mu_0 C_{ox}}{L} \left[(v_{GS} - V_t)v_{DS} - \frac{v_{DS}^2}{2} \right] (1 + \lambda v_{DS})$$

Where:

- μ_0 = zero field mobility ($\text{cm}^2/\text{volt}\cdot\text{sec}$)
- C_{ox} = gate oxide capacitance per unit area (F/cm^2)
- λ = channel-length modulation parameter (volts^{-1})
- V_{t0} = zero-bias threshold voltage
- γ = bulk threshold parameter ($\text{volts}^{-0.5}$)
- $2\Phi_f$ = strong inversion surface potential (volts)

Note: For p-channel MOSFETs, use n-channel equations with p-channel parameters and invert current.



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First-Order Large-Signal Model Equations (continued)

Saturation

$$i_D = \frac{W\mu_o C_{ox}}{L} \left[(v_{GS} - V_t) v_{DS(sat)} - \frac{v_{DS(sat)}^2}{2} \right] (1 + \lambda v_{DS})$$

$$i_D = \frac{W\mu_o C_{ox}}{2L} (v_{GS} - V_t)^2 (1 + \lambda v_{DS})$$

$$V_t = V_{t0} + \gamma \left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f} \right]$$

$$r_o = \frac{V_A}{I_D} \quad \text{where } I_D \text{ is the drain current without channel-length modulation taken into account.}$$

Where:

- μ_o = zero field mobility (cm²/volt·sec)
- C_{ox} = gate oxide capacitance per unit area (F/cm²)
- λ = channel-length modulation parameter (volts⁻¹)
- V_{t0} = zero-bias threshold voltage
- γ = bulk threshold parameter (volts^{-0.5})
- $2\Phi_f$ = strong inversion surface potential (volts)

Note: For p-channel MOSFETs, use n-channel equations with p-channel parameters and invert current.

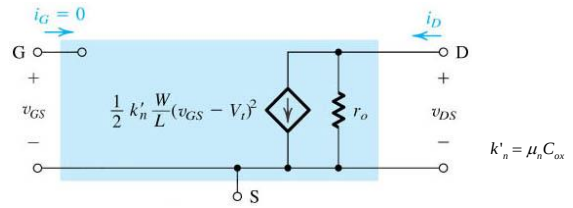


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Large-Signal Model

The information from the previous slide can be incorporated into a large-signal model with the following assumptions:

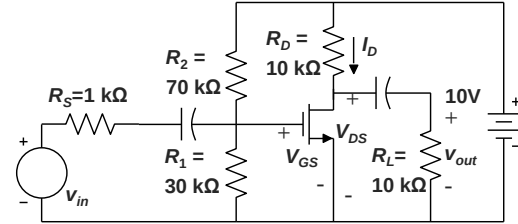
- Input impedance is ∞ .
- The transfer characteristic is modeled as a voltage-controlled current source.
- r_o is based on the early voltage, V_A (typically 2 to 50V, depending on the gate length and technology).



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Example: Large-Signal Model

Find the DC operating point of the NMOS transistor shown, assuming $k_n' = 25 \mu\text{A}/\text{V}^2$, $W/L = 10$, $V_{tN} = 1\text{V}$, and $\lambda = 0$.



Solution

$$V_{GG} = V_{GS} = \frac{R_1}{R_1 + R_2} V_{DD} = \frac{30\text{k}\Omega}{30\text{k}\Omega + 70\text{k}\Omega} (10\text{V})$$

$$\therefore V_{GG} = 3\text{V}$$

Assume the MOSFET is in saturation and find I_D :

$$I_D = \frac{Wk_n'}{2L} (V_{GS} - V_t)^2 = \frac{25 \cdot 10}{2} (3 - 1)^2 = 500 \mu\text{A}$$

Check to see if the transistor is in saturation:

$$V_{DS} = V_{DD} - I_D \cdot 10\text{k}\Omega = 5\text{V}$$

$$V_{DS} \geq V_{GS} - V_t \Rightarrow 5 > (3 - 1) \Rightarrow 5\text{V} > 2\text{V}$$

(Transistor in saturation: ✓)

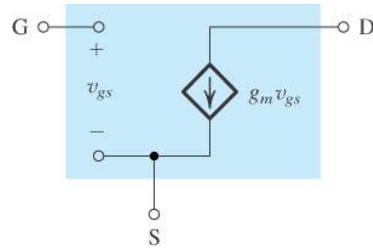
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Small-Signal Frequency-Independent Models

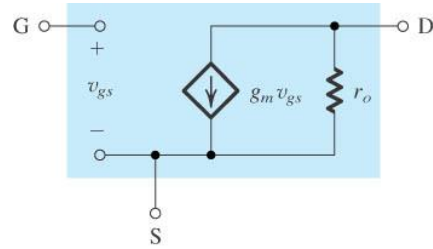
Two possible frequency-independent models are:

- With r_o
- Without r_o

The choice of one over the other depends on whether r_o significantly changes the resistance at the output.



(a)



(b)

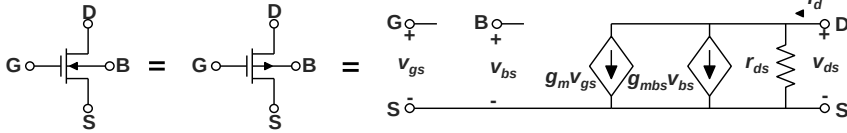
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Small-Signal Models

The small-signal model is essentially the linear part of a power-series expansion of the drain current dependence on gate, bulk, and drain voltages.

$$i_d \approx \left. \frac{di_D}{dv_{GS}} \right|_Q v_{gs} + \left. \frac{di_D}{dv_{BS}} \right|_Q v_{bs} + \left. \frac{di_D}{dv_{DS}} \right|_Q v_{ds} + \dots = g_m v_{gs} + g_{mbs} v_{bs} + g_{ds} v_{ds} + \dots$$

Schematic model:



Where:

$$g_m \equiv \left. \frac{di_D}{dv_{GS}} \right|_Q = \beta(v_{GS} - V_t) = \sqrt{2\beta I_D} \quad g_{ds} \equiv \left. \frac{di_D}{dv_{DS}} \right|_Q = \frac{\lambda i_D}{1 + \lambda v_{DS}} \approx \lambda i_D$$

$$g_{mbs} \equiv \left. \frac{\partial i_D}{\partial v_{BS}} \right|_Q = \left(\frac{\partial i_D}{\partial v_{GS}} \right) \left(\frac{\partial v_{GS}}{\partial v_{BS}} \right) \bigg|_Q = \left(-\frac{\partial i_D}{\partial V_t} \right) \left(\frac{\partial V_t}{\partial v_{BS}} \right) \bigg|_Q = \frac{g_m \gamma}{2\sqrt{2|\phi_f|} - v_{BS}} = \eta g_m$$

$$\beta \equiv \mu_0 C_{ox} \frac{W}{L} = k'_n \frac{W}{L}$$

Engineering approximation:

$$g_m \approx 10 g_{mbs} \approx 100 g_{ds}$$



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Small-Signal Models for the Nonsaturated Region

- In the nonsaturated (triode) region, g_m has to be calculated differently:

$$g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_Q = \frac{k'_n W_{DS}}{L} (1 + \lambda v_{DS}) \approx \left(\frac{k'_n W}{L} \right) v_{DS}$$

$$g_{mbs} = \left. \frac{\partial i_D}{\partial v_{BS}} \right|_Q = \frac{k'_n W \lambda v_{DS}}{2L \sqrt{2\phi_f - v_{BS}}}$$

$$g_{ds} = \left. \frac{\partial i_D}{\partial v_{DS}} \right|_Q = \frac{k'_n W}{L} (v_{GS} - V_t - v_{DS}) (1 + \lambda v_{DS}) + \frac{I_D \lambda}{1 + \lambda v_{DS}} \approx \frac{k'_n W}{L} (v_{GS} - V_t - v_{DS})$$

- The most obvious difference is that all of the transconductances are now functions of v_{DS} not i_D (as long i_D is small enough for $V_{DS} < V_{DSat}$).



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Frequency-Dependent Small-Signal Models

Inductance effects for devices in ICs are usually smaller than capacitance effects and are usually neglected in the small-signal model.

There are two types of capacitance regarding MOSFETs:

- **Charge storage:** Caused by the gate and channel acting as capacitor plates with the gate oxide acting as the dielectric.
- **Depletion:** Caused by the depletion layer formed by reverse-biased pn junctions.

Some models for very high frequencies also include delay effects.



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MOSFET Capacitances

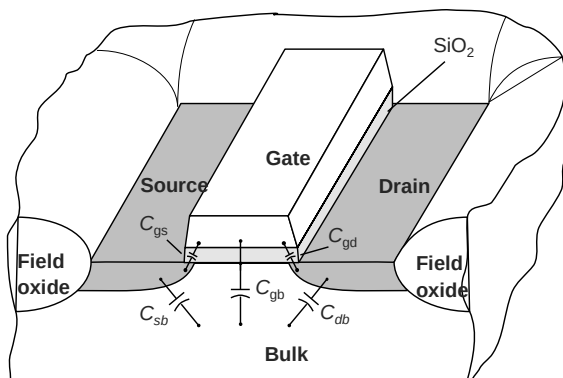
Five primary parasitic capacitances must be accounted for in the small-signal frequency-dependent model:

- Charge storage

- C_{gs}
- C_{gd}

- Depletion

- C_{gb}
- C_{sb}
- C_{db}



Note: For the designer, usually, the total of C_{GS} and C_{GD} is the most important. The five capacitances cannot be easily separated, which is a problem for parameter extraction. Consequently, special test structures are often used.

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MOSFET Charge-Storage Capacitances

In the triode region with the channel of uniform depth, the total capacitance can be split between source and drain:

$$C_{gs} = C_{gd} = \frac{1}{2} WLC_{ox}$$

In the saturation region, the channel at the drain is pinched off so that most of the capacitance is assigned to the source and none to the drain:

$$C_{gs} = \frac{2}{3} WLC_{ox} \quad C_{gd} = 0$$

When the MOSFET is cut off, there is no channel, and therefore no capacitance between the gate and the source or drain. There is still a capacitance between the gate and substrate:

$$C_{gs} = C_{gd} = 0 \quad C_{gb} \approx WLC_{depletion}$$

For more accuracy, an additional capacitance should be added to the previous values. This capacitance is caused because the source and drain diffusions exist slightly under the gate oxide. Typically, $L_{ov} = 0.05$ to $0.1L$.

$$C_{ov} = WL_{ov}C_{ox}$$



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MOSFET Depletion Capacitances

The depletion capacitances can be found using the following formulas:

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{V_{SB}}{V_0}}} \quad C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{V_{DB}}{V_0}}}$$

Where:

- C_{sb0} and C_{db0} are the capacitances at zero reverse-bias voltage.
- V_{SB} and V_{DB} are the magnitudes of the reverse-bias voltages.
- V_0 is the junction built-in voltage (typically, approximately 0.6V to 0.8V).



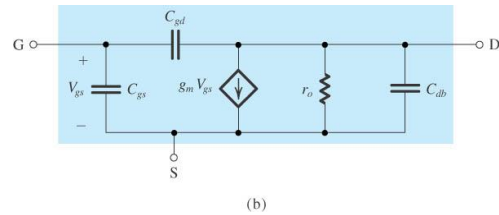
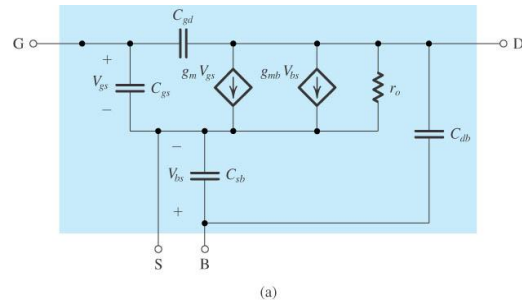
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Equivalent Small-Signal Models

Putting together the small-signal model with the capacitors results in the equivalent small-signal frequency-dependent model shown in figure (a).

In the common case when the source and bulk are tied together, the slightly simpler model shown in figure (b) can be used.

Note: This represents the so-called *intrinsic* MOSFET device. Series resistors at each terminal should be added for representing the complete device.



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Summary: MOSFET Models

Models

- Large signal
- Small signal
 - Frequency independent
 - Active mode
 - Saturation mode
 - Frequency dependent

MOSFET capacitances

- Charge storage
- Depletion



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Introduction: CMOS Fabrication

- **Wafer preparation:** Processing the raw semiconductor material to be of extremely high quality with few defects and grown as a single crystal.
- **Photolithography:** Transferring the circuit layout to the wafer by the use of masks to reveal some parts of the wafer and hide others.
- **Oxidation:** Forming a very uniform oxide layer on the silicon, which is used to insulate process layers from one another.
- **Ion implantation/diffusion:** Doping specific parts of the wafer to be an n-type or p-type semiconductor.
- **Deposition and etching:** Depositing materials on the wafer and removing the unwanted material.



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Wafer Preparation: Materials

The wafer can be made of any of a number of different semiconductors:

- Silicon (Si)
- Gallium arsenide (GaAs)
- Silicon germanium (SiGe)
- Other exotic materials

The most common base material is silicon.



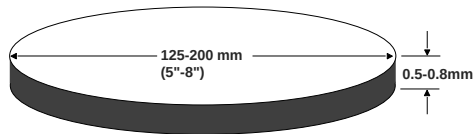
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Wafer Preparation: Czochralski Method

The silicon is grown by the Czochralski method:

- Starts with a single silicon seed.
- The seed is immersed in molten silicon and slowly rotated and pulled.
- The result is an ingot of single-crystal silicon with few defects.

The ingot is sliced into thin wafers that form the starting point for IC fabrication:



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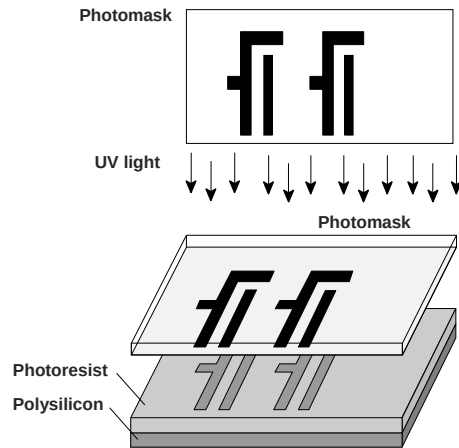
Photolithography

- Photolithography is the process of using light to transfer an image to a surface (in this case, the silicon wafer) by coating the surface of the wafer with an ultraviolet-light-sensitive material called *photoresist*.
- The photoresist is then developed with a chemical that removes the UV-processed areas, leaving a replica of the pattern on the surface of the wafer.
- Depending on the type of photoresist used, either the photoresist exposed to UV radiation is removed (positive photoresist), or the areas not exposed to UV radiation are removed (negative photoresist).



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Illustration of Photolithography

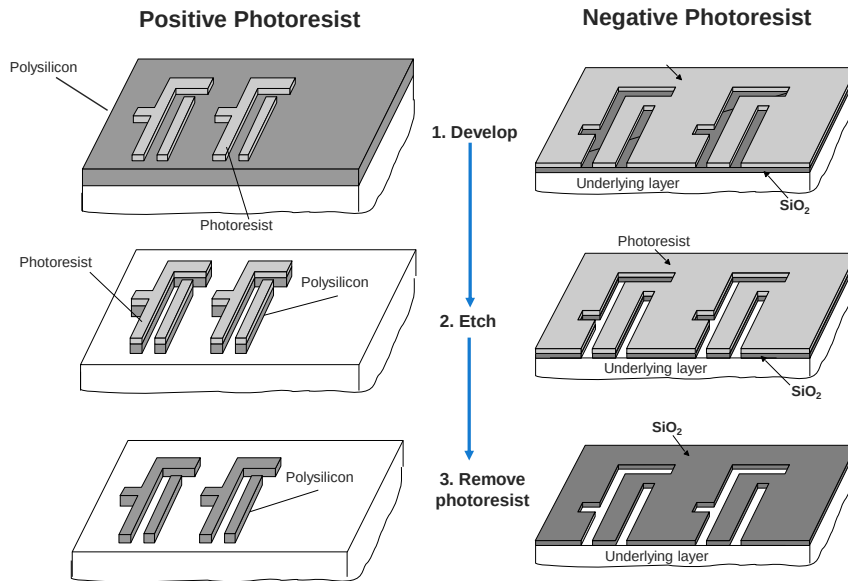


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Positive and Negative Photoresist



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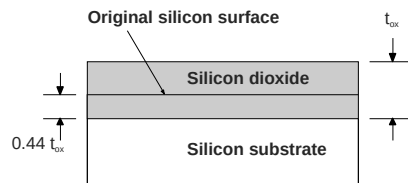


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Oxidation

Oxidation is the process by which a layer of silicon dioxide is grown on the surface of a silicon wafer:

- It protects the underlying material from contamination.
- It provides isolation between two layers.
- Its thicknesses can vary between $10\text{\AA}$ to over $1000\text{\AA}$.
- MOS transistor oxide is the thinnest oxide layer. Field oxide is much thicker.

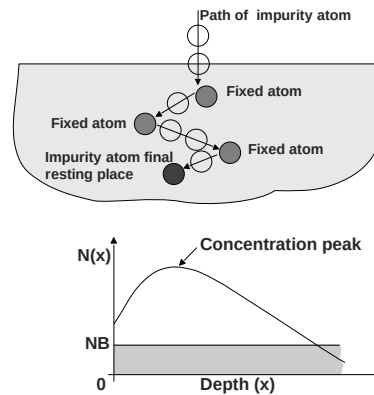


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Ion Implantation

Ion implantation is the process by which impurity ions are accelerated to a high velocity and physically lodged into the target material:

- Annealing is required to activate the impurity atoms and repair the physical damage to the crystal lattice. This step is done at 500° to 800° C.
- Ion implantation is a lower-temperature process compared to diffusion and enables more shallow doping profiles, which are required for submicron devices.
- The process can be implanted through surface layers. Thus, it is useful for field-threshold adjustment.
- Ion implantation can achieve unique doping profiles, such as buried concentration peak.

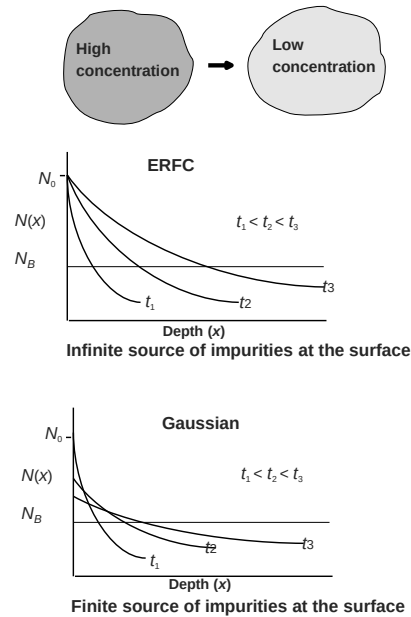


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Diffusion

Diffusion is the movement of impurity atoms at the surface of the silicon into the bulk of the silicon:

- Diffusion is always in the direction from higher concentration to lower concentration.
- Diffusion is typically done at high temperatures of 800° to 1400° C.



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Deposition

Deposition is the means by which various materials are deposited on the silicon wafer:

- Examples
 - Silicon nitride (Si_3N_4)
 - Silicon dioxide (SiO_2)
 - Aluminum
 - Polysilicon
- There are various ways to deposit a material on a substrate:
 - Chemical-vapor deposition (CVD)
 - Low-pressure chemical-vapor deposition (LPCVD)
 - Plasma-assisted chemical-vapor deposition (PECVD)
 - Sputter deposition

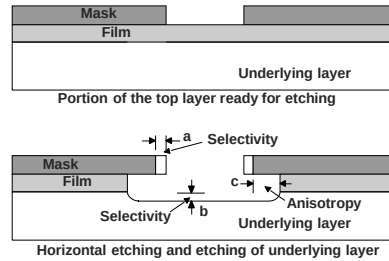


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Etching

Etching is the process of selectively removing a layer of material:

- When etching is performed, the etchant might remove portions or all of the
 - Desired material
 - Underlying layer
 - Masking layer
- Important considerations
 - Anisotropy (A)
 - Selectivity (S)
 - Ideally, $A = 1$ and $S = \infty$
- There are two types of etches:
 - Wet etch – uses chemicals
 - Dry etch – uses chemically active ionized gases



$$A = 1 - \frac{(\text{lateral_etch_rate})}{(\text{vertical_etch_rate})}$$

$$S = \frac{(\text{film_etch_rate})}{(\text{mask_etch_rate})}$$

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Process and Device Variability

All process parameters, such as temperatures used, doping levels, and so forth, vary slightly:

- There are global variations (between lots, between wafers in a lot, and so forth.) and local variations (for example, between two devices close together).
- In the past people could assume that, *within* a wafer, the parameters would not vary much, but there were significant variations from lot to lot. However, modern wafers are so large that across one wafer there are also significant parameter gradients.
- Local variations are due to the atomic nature of the materials plus process imperfections.

Layer thickness and doping levels vary; therefore, the resistivity of the resistor layer varies or unit capacitances, also transistor threshold voltage, breakdown voltages, and so forth, vary as well.



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Process and Device Variability (continued)

The length and width of components vary:

- Length and width are not 100% as defined by the mask.
- The mask is not 100% as defined by the designer or the software.
- Additional variations in resistances, and so forth.

Even devices that are close together and drawn in the mask, in the same way, do not have 100% of the same characteristics:

- Matching is limited, which is one major problem in any analog design because systematic variations are usually easier to compensate for than random imperfections.
- The most critical is limited matching for threshold voltage V_{TO} of MOS transistors.



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Summary: CMOS Fabrication

- CMOS fabrication steps
 - **Wafer preparation:** Preparing the silicon for use.
 - **Photolithography:** Transferring the circuit layout to the wafer:
 - **Oxidation:** Forming a very uniform oxide layer on the silicon.
 - **Ion implantation/diffusion:** Doping specific parts of the wafer.
 - **Deposition and etching:** Depositing materials on the wafer and removing the unwanted material.
- Processing variability



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Introduction: CMOS Switching Behavior

The switching performance of a CMOS logic gate can be specified with the following parameters:

- Noise margin
- Propagation delay, rise time, and fall time
- Power dissipation
- Delay-power product
- Fan-in/fanout

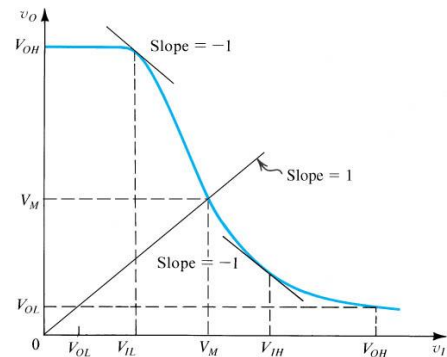


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Noise Margin

Definition: Parameter that indicates the susceptibility of a circuit to noise.

- The static operation of a gate is given by its voltage transfer characteristic (VTC).
- The VTC defines the output voltage as a function of the input voltage.
- The VTC has five parameters:
 - V_{OH}
 - V_{OL}
 - V_{IH} (defined where slope = -1)
 - V_{IL} (defined where the slope = -1)
 - V_M or V_{th} (defined where $v_o = v_i$)
- Noise margin can be defined as
 - $N_{MH} = V_{OH} - V_{IH}$
 - $N_{ML} = V_{IL} - V_{OL}$
- The noise margin measures the robustness of a logic family. The smaller the margins, the more susceptible the family is to noise.



Inverter curve showing noise margin measurements.

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Propagation Delay

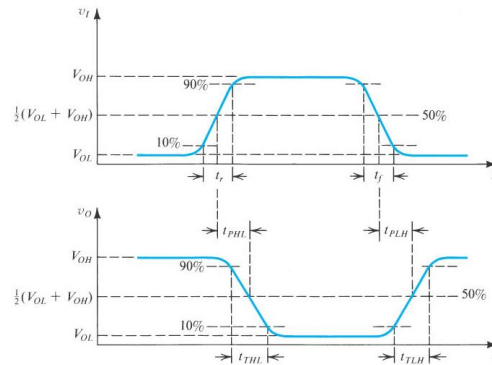
Definition: The time it takes for a signal at the input to affect the output signal.

Propagation delay characterizes the dynamic performance of a logic circuit:

- The overall propagation delay is defined as the average of the two specific delays:

$$t_{PD} = \frac{1}{2}(t_{PLH} + t_{PHL})$$

- Shorter delays equal faster overall operation.
- Rise and fall times are asymmetric due to different mobility, thresholds, and sizing of NMOS and PMOS transistors.



Definition of low-to-high and high-to-low propagation delay.

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Power Dissipation

Definition: The amount of power dissipated in a circuit.

Low-power dissipation is important:

- Reduces heat
- Reduces power requirements (increases battery life)
- Increases circuit density

The two types are

- **Static:** Power that is dissipated when not switching.
- **Dynamic:** Power that is dissipated during switching.

Dynamic power dissipation can be calculated as $P_D = fCV_{DD}^2$.

CMOS advantage: Per gate, static power dissipation is virtually zero. However, for millions of gates, the power dissipation might not remain negligible.



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Delay-Power Product

Definition: A figure of merit that is used to compare different logic families:

- High-speed and low-power dissipation are usually in conflict. Lower propagation delays usually require more power.
- The delay-power product is a figure of merit that can be used to compare various logic technologies:

$$DP = P_D t_p$$



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Fan-In and Fanout

- **Fan-in:** The number of inputs on a gate.
- **Fanout:** The maximum number of gates of similar size that the gate can drive while remaining within specifications.

CMOS: Because static current is low, even a small gate can drive many others, but each additional gate adds capacitance and reduces overall speed.



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Summary: CMOS Switching Behavior

CMOS switching parameters

- Noise margin
- Propagation delay, rise time, and fall time
- Power dissipation
- Delay-power product
- Fan-in/fanout



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Introduction: CMOS Logic Structures

Logic gate circuits

- Basic structure
 - Symbols and conventions
 - PUN (Pull-up network)/PDN (Pull-down network)
- Combinatorial circuits
 - Inverter
 - NOR gate
 - NAND gate
 - XOR gate
 - More complex gates
- Sequential circuits
 - Latches
 - Flip-flops
- Non-CMOS logic circuits
 - Pass-transistor logic circuits
 - NMOS and CMOS switches
 - Dynamic logic

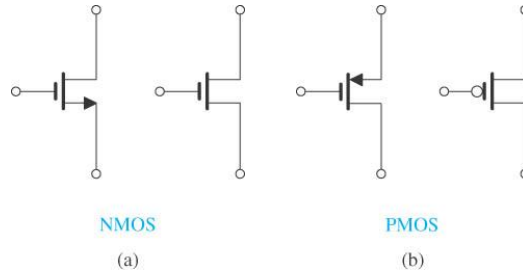


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Circuit Symbols and Boolean Conventions

Introduction of symbols and conventions

- To reduce schematic complexity, a simpler MOSFET symbol is used for logic gates as shown.
- Boolean conventions:
 - OR: $Y = A + B$
 - AND: $Y = AB$



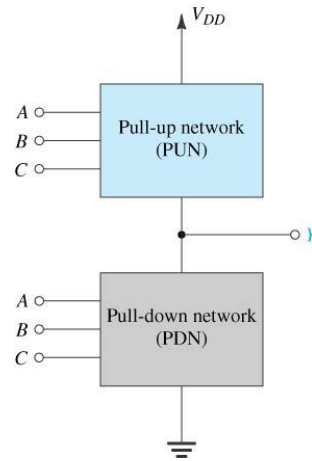
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Basic Structure of a CMOS Logic Gate

All CMOS gates are an extension of the basic inverter. The CMOS logic gate can look at any other CMOS gate as two networks:

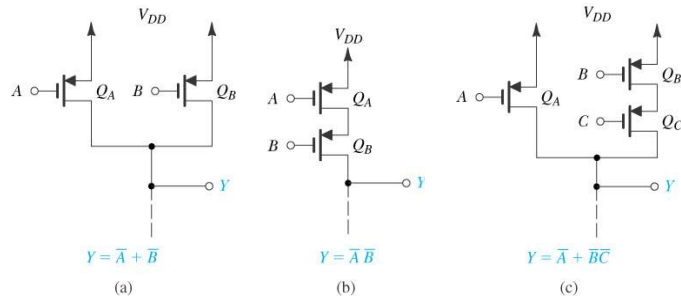
- **PUN (pull-up network):** Consists of PMOS devices and is connected to the upper voltage rail (usually V_{DD}).
- **PDN (pull-down network):** Consists of NMOS devices and is connected to the lower voltage rail (usually ground).

Because the PDN consists of NMOS devices, it is activated when the inputs are high. Likewise, the PUN is activated by low signals.



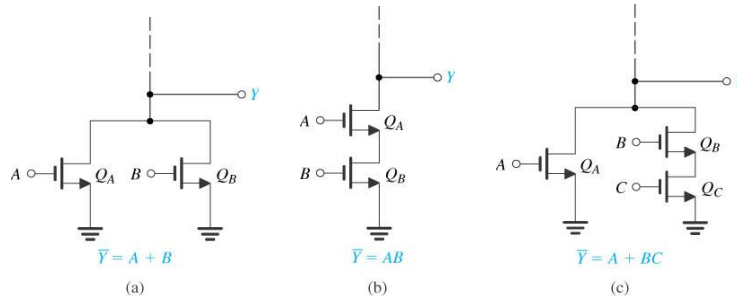
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Example: Pull-Up Networks



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Example: Pull-Down Networks



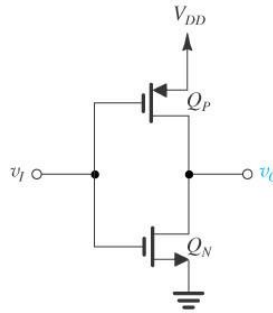
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Basic Circuits: CMOS Inverter

Inverter: $Y = \bar{A}$

Design Procedure

- **PUN:** Notice that when v_O is high, the input is low. The PUN requires only one PMOS device.
- **PDN:** Notice that when v_O is low the input is high. Therefore, the PDN consists of only one NMOS device.



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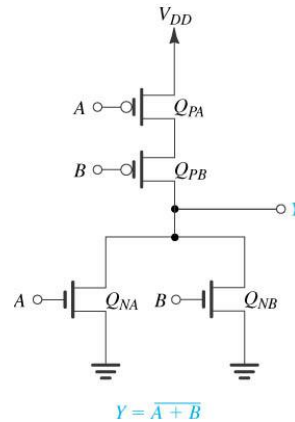
Basic Circuits: CMOS NOR Gate

NOR Gates: $Y = \overline{A + B} = \overline{A} \cdot \overline{B}$

Design Procedure

- **PUN:** Using the second form of the equation, notice that A and B must both be low to make Y high. Therefore, the PUN consists of two series of PMOS devices.
- **PDN:** Notice that Y is low when either A or B is high. Therefore, the PDN consists of two parallel NMOS devices.

Note: You should avoid having too many PMOS devices in a series.



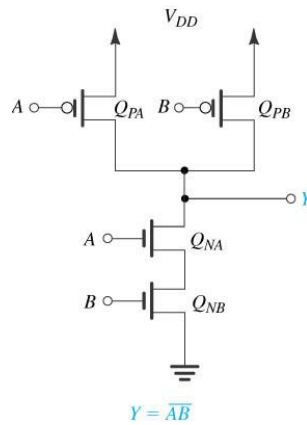
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Basic Circuits: CMOS NAND Gate

NAND Gates: $Y = \overline{AB} = \overline{A} + \overline{B}$

Design Procedure

- **PUN:** Using the second form of the equation, notice that either A or B must be low to make Y high. Therefore, the PUN consists of two parallel PMOS devices.
- **PDN:** Notice that Y is low when both A and B are high. Therefore, the PDN consists of two series of NMOS devices.



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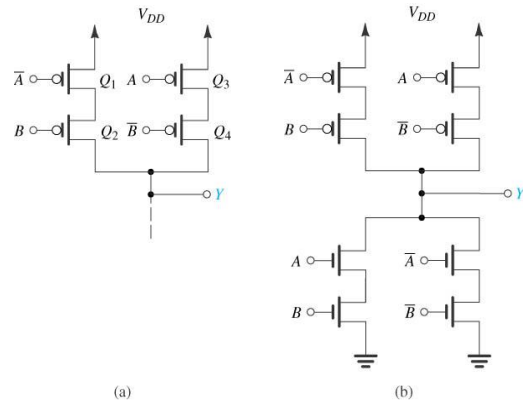
Basic Circuits: CMOS XOR Gate

XOR Gates: $Y = A \cdot \bar{B} + \bar{A} \cdot B$

Design Procedure

- Because Y is given, it is easier to design the PUN first. The design for the PUN is shown.
- To generate the PDN, get an expression for Y and then synthesize:

$$\begin{aligned}\bar{Y} &= \overline{A \cdot \bar{B} + \bar{A} \cdot B} \\ &= \overline{A \cdot \bar{B}} \cdot \overline{\bar{A} \cdot B} \\ &= (\bar{A} + B) \cdot (A + \bar{B}) \\ &= A \cdot B + \bar{A} \cdot \bar{B}\end{aligned}$$



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Basic Circuits: More Complex Gates

General PUP/PDN approach: $Y = \overline{A \cdot (B + C \cdot D)}$

Find expressions for Y and Y using

DeMorgan's law:

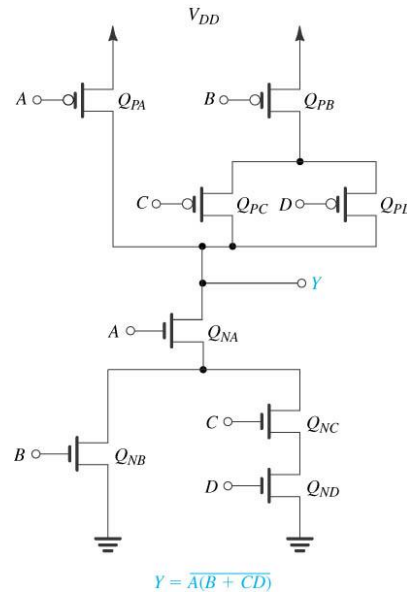
$$\bar{Y} = A \cdot (B + C \cdot D)$$

Design Procedure

- PDN: Notice that the PDN network can be created by inspection.
- PUN: Using DeMorgan's law:

$$\begin{aligned} Y &= \overline{A \cdot (B + C \cdot D)} \\ &= \bar{A} + \bar{B + C \cdot D} \\ &= \bar{A} + \bar{B} \cdot \overline{C \cdot D} \\ &= \bar{A} + \bar{B} \cdot (\bar{C} + \bar{D}) \end{aligned}$$

- Now the PUN can be created.



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Sequential Logic

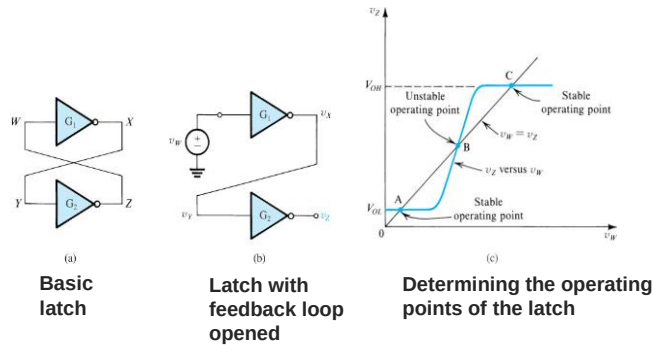
- *Combinatorial logic* changes whenever the inputs change. *Sequential logic* only changes at discrete times, usually through the use of a periodic signal called a clock.
- *Sequential logic* is useful for applications where information must be stored. The most common circuits are latches and flip-flops, which are capable of storing one bit of information until commanded to change.



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Latches

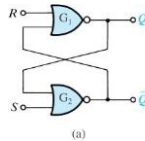
- The latch is the basic memory element. The inverters shown are connected in a positive feedback loop so that the output ends up at either V_{OH} or V_{OL} because any other point is unstable.
- Based on external excitation, the latch achieves one of the two states and then holds that output indefinitely. Without inputs, it is not particularly useful, but the structure forms the basis of something far more useful flip-flops.



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Setting and Resetting Flip-Flops

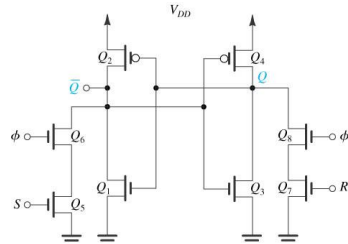
- The simplest flip-flop (the set/reset – SR– flip-flop) uses the positive feedback principle of the latch to hold data. A schematic and truth table are shown.
- Improving the design means incorporating the ability to also clock the flip-flop, as shown in the right-most figure. The clock signal is denoted by ϕ .



SR flip-flop

R	S	Q_{n+1}
0	0	Q_n
0	1	1
1	0	0
1	1	Not used

SR flip-flop
truth table

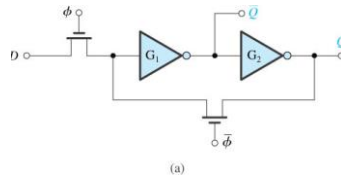


CMOS implementation of
clocked SR flip-flop

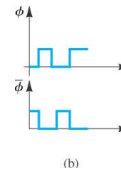
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Data Flip-Flops

- The data (D) flip-flop is an example of an edge-triggered design. When the clock is low, the flip-flop holds the output data. When the clock goes high, the output changes state to whatever the input signal was just prior to the rising edge of the clock.
- In the simple implementation of the D flip-flop shown, notice that the two clock phases must be nonoverlapping. In other words, proper operation requires that the two clock phases are never high at the same time.



Circuit with two-phase nonoverlapping clock

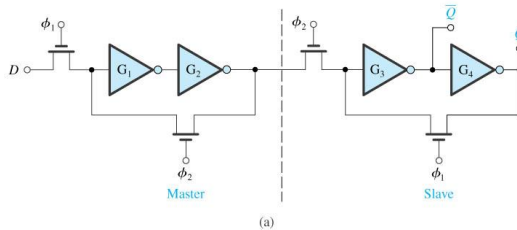


Waveforms of D flip-flop circuit

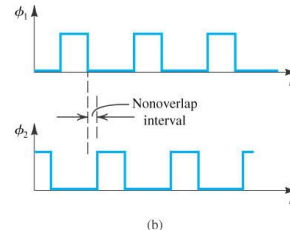
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Data Flip-Flops (continued)

The previous design has one flaw: When ϕ is high, the output follows the input which can create problems in some designs. The following design, called a *master-slave D flip-flop*, fixes that flaw. The switches can be, and usually are, implemented with CMOS transmission gates.



Master-slave D flip-flop



Waveforms of the two-phase nonoverlapping clock required

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Nonstandard Logic Structures

Nonstandard logic structures are gates that are not considered true CMOS. The reasons for choosing one include:

- The need for very high density.
- The need for very high speed or very low power.

Some logic types are particularly good for specific circuits. An example of this is the use of pass-transistor logic for multiplexers.

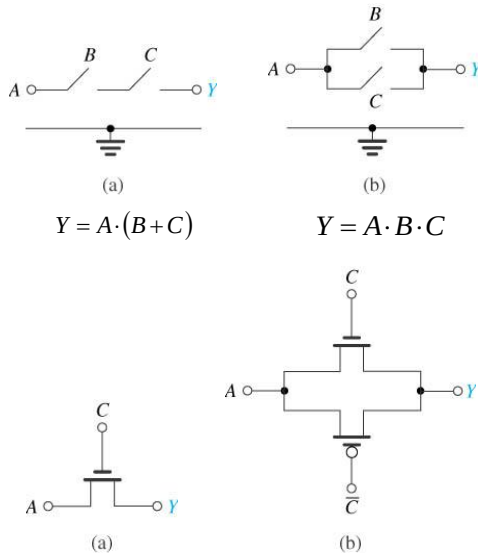


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Pass-Transistor Logic Circuits

Pass-transistor logic circuits (PTL) use CMOS transmission gates in series and/or parallel to realize various gates:

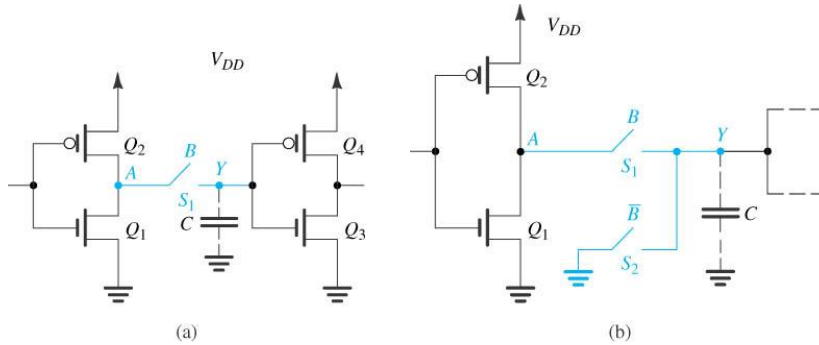
- Depending on the application, one of two types of transmission gate can be used:
 - Single NMOS T-gate
 - CMOS T-gate
- In the CMOS T-gate, notice the need to provide complementary control inputs.
- PTL has no buffering, that is, the load current has to be delivered from the driving source and the output swing does not recover (at best equal to input swing).



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Low-Resistance PTL Paths

Note: To satisfy static logic requirements *every circuit node must have at all times a low-resistance path to ground or V_{DD} available*. Otherwise, a dynamic logic circuit is created that needs constant refreshing of the gate charge similar to DRAM bits.



With S_1 open, node Y floats and output of Q3/Q4 is indeterminate.

Providing a complementary switch S_2 solves floating node problem.



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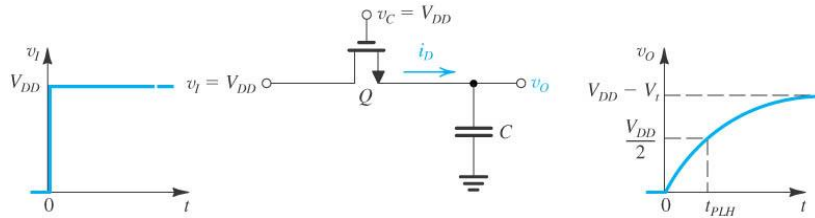
NMOS Switches

Advantages

- Small area
- Small node capacitances

Disadvantages

- Slow rise and fall times.
- Output voltage is not V_{DD} but $V_{DD}-V_t$.
- Reduced gate noise immunity.
- Can have static current/power dissipation.

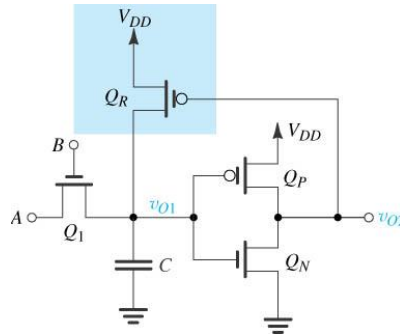


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Modified NMOS Switches

The following are two possible improvements to NMOS switches:

- **Circuit-based approach:** Use of a second transistor to restore the level to V_{DD} .
- **Process technology approach:** Use devices that have zero threshold voltage. Fabricated by ion implantation and called *natural devices*.



Note: These devices have higher than normal leakage because when off $V_{gs} = V_t$.

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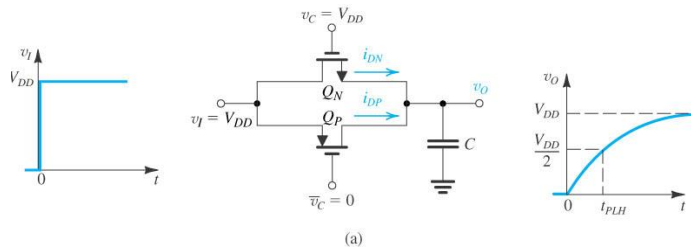
CMOS Switches

Advantages

- Good static and dynamic performance
- Bidirectional current flow
- Good for analog applications also

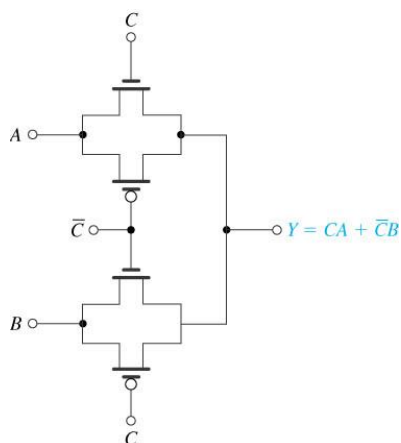
Disadvantages

- Larger than NMOS switch
- Need for additional inverted drive

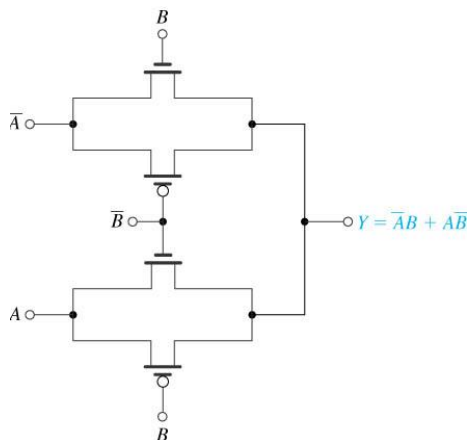


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CMOS Switch Examples



2:1 Multiplexer



XOR

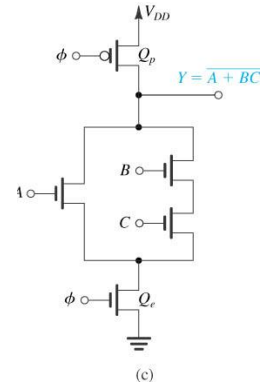
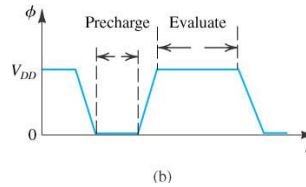
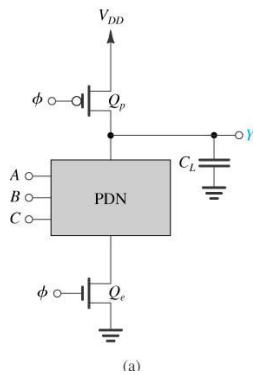
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Dynamic Logic Gates

- Store signals on parasitic capacitances on open nodes.
- Require a clock to periodically refresh the data.
- Have a low-device count, higher speed, and lower power at the expense of increased complexity and loss of robustness.



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Dynamic Logic Gates (continued)

- When Φ is low, the storage node with load C_L is charged to V_{DD} . Inputs are allowed to change and settle.
- When Φ is high, the PDN network evaluates the inputs and either conduct ($Y=0$) or does not ($Y=V_{DD}$).
- The output is valid only when the clock is high (evaluate), because Y is always forced to V_{DD} when the clock is low (precharge).

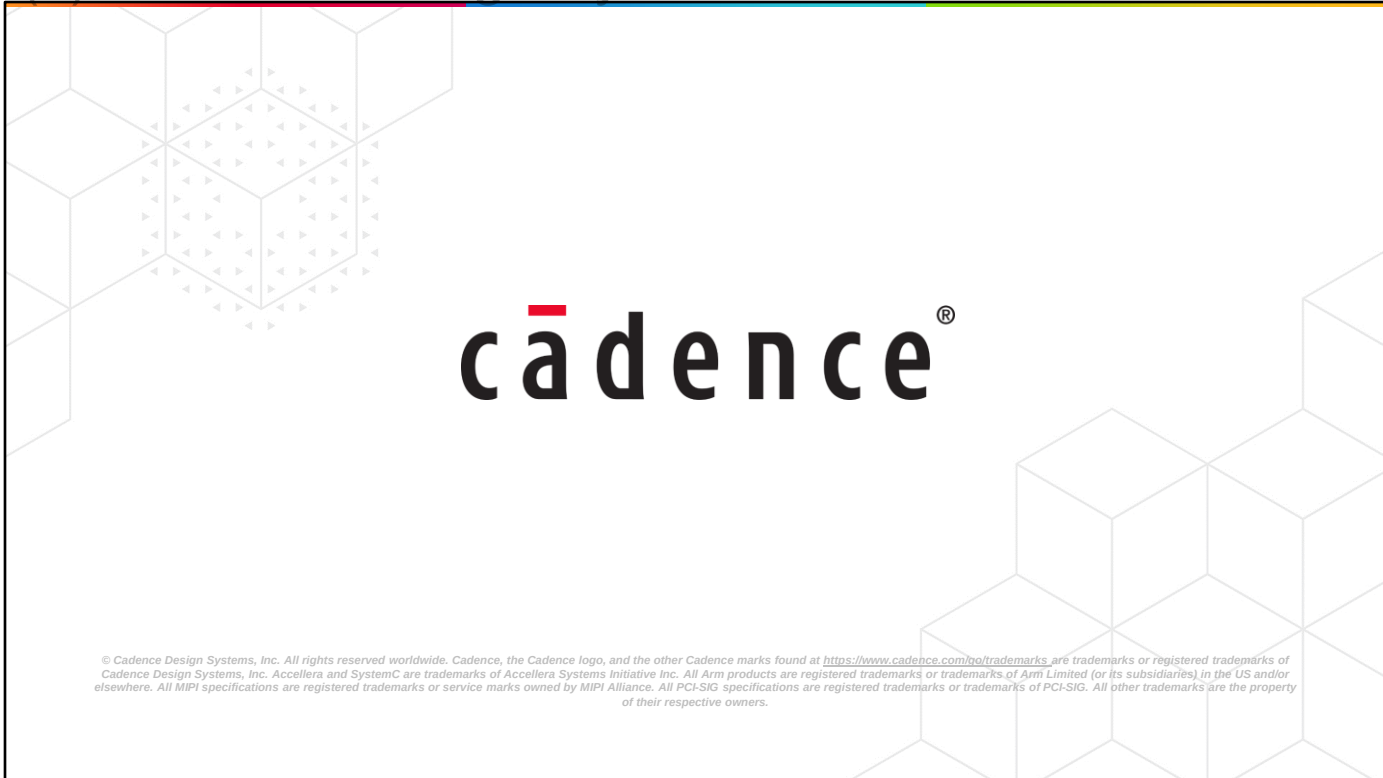
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Summary: Logic Gate Circuits

- Basic logic gate structure
 - Symbols and conventions
 - PUN (pull-up network)/PDN (pull-down network)
- Standard CMOS circuits
 - Combinatorial circuits (NAND, NOR, and so forth)
 - Sequential circuits (flip-flops)
- Other logic structures
- Pass-transistor logic circuits
- NMOS and CMOS switches
- Dynamic logic



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